

GEUVADIS Gene Expression GREML Workflow Handoff

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This document is a knowledge-transfer guide for the GEUVADIS gene expression heritability workflow in this repository. It is written for the next person who needs to understand, rerun, audit, or extend the analysis after the original workflow owner leaves the project.

The workflow estimates gene-level SNP heritability of RNA expression phenotypes using GEUVADIS RNA-seq data, 1000 Genomes genotype data, GTEx-inspired expression processing, and GCTA GREML. The repository also contains downstream diagnostic analyses to compare expression choices, normalization choices, SNP sets, and expression distribution properties.

1. Scientific Question

The central question is:

How much of gene expression variation across European GEUVADIS individuals can be explained by genotype-derived relatedness, and how sensitive are those heritability estimates to expression quantification, normalization, SNP set, and expression distribution shape?

Operationally, each gene is treated as a quantitative phenotype. GCTA GREML is used to estimate the fraction of phenotypic variance explained by a genomic relationship matrix:

$$h^2_{\text{GREML}} = V(G) / V_p$$

where 'V(G)' is the additive genetic variance component estimated from the GRM and 'Vp' is total phenotypic variance after the model accounts for covariates.

The workflow asks this question under multiple preprocessing choices:

- * SNP set: all available SNPs in the genotype prefix versus HapMap3/no-MHC SNPs.
- * Expression source: TPM-derived expression versus TMM-normalized count-derived expression.
- * Phenotype normalization: raw expression scale versus inverse-normal transformed expression.
- * Covariates: genotype PCs and PEER factors.

The broader scientific motivation is that expression heritability estimates can be sensitive to the phenotype scale. Low expression, zero inflation, skewed expression distributions, and different normalization choices can all change how GREML partitions variance.

2. Previous Research And What This Workflow Mimics

GEUVADIS and 1000 Genomes

GEUVADIS generated RNA-seq data for lymphoblastoid cell lines from 1000 Genomes individuals. The public GEUVADIS collection contains mRNA and small RNA sequencing from over 460 samples in CEU, FIN, GBR, TSI, and YRI. This workflow focuses on the European-ancestry groups used in many GEUVADIS analyses:

- * British: GBR
- * Finnish: FIN
- * Tuscan: TSI
- * Utah residents with Northern and Western European ancestry: CEU

The project uses GEUVADIS expression data because it has both population-scale RNA-seq and matched 1000 Genomes genotype data. That makes it a good test case for expression heritability: for the same individuals, we can construct gene expression phenotypes and genotype-derived GRMs.

Hernandez et al.-style expression heritability framing

Hernandez et al. studied gene expression heritability in GEUVADIS and emphasized how variant frequency classes, rare variants, and expression architecture affect estimated cis heritability. In their GEUVADIS European analysis, they started from European individuals and removed cryptically related samples using 1000 Genomes relatedness/identity-by-state information, ending with an unrelated EUR set.

This workflow mimics the Hernandez-style sample logic by:

- * Starting from GEUVADIS EUR metadata.
- * Mapping RNA-seq run IDs to 1000 Genomes genotype IDs.
- * Optionally intersecting the mapped EUR sample set with 'unrelated_keep_file'.
- * Using that sample set consistently for GRM, PCA, phenotype prep, and GREML.

Important distinction:

```
Hernandez-style pruning = external 1000G unrelated sample keep list  
not necessarily GCTA --grm-cutoff 0.025
```

The workflow now exposes this as:

```
params.unrelated_keep_file
```

GTEx-style expression phenotype processing

The workflow intentionally follows GTEx-like expression processing choices:

- * Keep genes with TPM ≥ 0.1 in at least 20% of samples.
- * Keep genes with counts ≥ 6 in at least 20% of samples.
- * Use TMM normalization for counts.
- * Use inverse-normal transformation per gene for normalized expression phenotypes.
- * Use genotype PCs and PEER factors as covariates.
- * Use a sample-size-based rule for PEER factors:
'15, 30, 45, 60' for increasing sample-size bins.

GTEx used these types of steps to make expression phenotypes more robust for linear modeling and eQTL mapping. This workflow reuses that logic before GREML.

TMM normalization rationale

TMM, the trimmed mean of M-values method from Robinson and Oshlack, was developed because simple library-size scaling can be biased when RNA composition differs between samples. If a sample has a few highly expressed genes, those genes take up sequencing "real estate" and can make all other genes appear lower by simple proportional scaling. TMM estimates sample-specific scaling factors after trimming extreme log-fold changes and intensity values, producing effective library sizes for cross-sample comparison.

In this pipeline, when 'expression_source=tmm', counts are converted to TMM-normalized CPM:

```
DGEList(counts)  
calcNormFactors(method = "TMM")  
cpm(normalized.lib.sizes = TRUE, log = FALSE, prior.count = 0)
```

GCTA GREML rationale

GCTA GREML uses a genomic relationship matrix as the covariance structure for random additive genetic effects. If genetically more similar individuals also have more similar expression for a gene, GREML attributes some expression variance to genotype. The output 'V(G)/Vp' is interpreted as the SNP heritability captured by the SNPs used to build the GRM.

In this workflow, the GRM is built once per run setting and the same GRM is used across genes. Each gene is then selected from a multi-phenotype file using GCTA's '--mpheno' option.

3. Repository Layout

Primary files:

```
README.md
run_greml.sh
nf/main.nf
nf/nextflow.config
nf/bin/prepare_phenotypes.R
scripts/
docs/
```

Important documentation already in the repo:

```
docs/how_to_run_greml_nextflow.md
docs/how_to_prepare_gtex_like_phenotypes.md
docs/how_to_run_downstream_analysis.md
docs/downstream_analysis_methods.md
```

Important analysis scripts:

```
scripts/plot_greml_h2_summary_boxplot.R
scripts/pairwise_h2_comparisons.R
scripts/tmm_expression_h2_deciles.R
scripts/tmm_raw_skewness_analysis.R
scripts/tpm_tmm_peer_correlations.R
scripts/downstream_h2_regulatory_repeat_analysis.R
```

4. Main HPC Paths

Project root on HPC:

```
/gpfs/data/mostafavilab/sool/analysis/GeneExpression/20260428_GE_GEUVDIS_v2/GeneExpression
```

Run directory root:

```
/gpfs/data/mostafavilab/sool/analysis/GeneExpression/20260428_GE_GEUVDIS_v2/GeneExpression/runs
```

Nextflow scratch/work root:

```
/gpfs/scratch/sl8085/nextflow_work/greml_production
```

RNA-seq matrices from the previous Salmon workflow:

```
/gpfs/data/mostafavilab/sool/analysis/GeneExpression/20260125_salmon_nextflow/results/matrices/gene_tpm.
tsv
/gpfs/data/mostafavilab/sool/analysis/GeneExpression/20260125_salmon_nextflow/results/matrices/gene_coun
ts.tsv
```

Default genotype prefix:

```
/gpfs/data/mostafavilab/shared_data/LDSC_data/1000G_EUR_Phase3_plink/1000G_EUR.QC.22
```

This resolves to:

```
1000G_EUR.QC.22.bed
1000G_EUR.QC.22.bim
1000G_EUR.QC.22.fam
```

The current prefix ends with '.22'. A successor should verify whether this is intentionally chromosome 22 only or whether the intended production GRM should be built from genome-wide merged chromosomes. This is one of the most important handoff checks.

GCTA executable:

```
/gpfs/data/mostafavilab/programs/GCTA/gcta-1.95.0-linux-kernel-3-x86_64/squashfs-root/AppRun
```

HM3/no-MHC SNP list:

```
/gpfs/data/mostafavilab/shared_data/LDSC_data/hm3_no_mhc_snps.txt
```

GEUVADIS SDRF metadata:

```
https://www.ebi.ac.uk/arrayexpress/files/E-GEUV-1/E-GEUV-1.sdrf.txt
```

5. Upstream RNA-seq Quantification

The previous Nextflow workflow quantified GEUVADIS RNA-seq with Salmon using GRCh37/hg19 Ensembl 75 transcriptome references:

```
Homo_sapiens.GRCh37.75.gtf.gz  
Homo_sapiens.GRCh37.75.cdna.all.fa.gz
```

It did the following:

1. Build a Salmon transcriptome index from the cDNA fasta.
2. Build a 'tx2gene.tsv' map from the GTF.
3. Run 'salmon quant' per GEUVADIS FASTQ using:

```
--validateMappings  
--seqBias  
--gcBias  
--geneMap tx2gene.tsv
```

4. Merge per-sample 'quant.genes.sf' into:

```
gene_counts.tsv  
gene_tpm.tsv
```

The Salmon workflow itself produced TPM and estimated counts. It did not create TMM. TMM is calculated later during phenotype preparation from the gene-level count matrix.

6. Main Nextflow Workflow Logic

The main workflow is in:

```
nf/main.nf
```

The main configuration is in:

```
nf/nextflow.config
```

The Slurm driver is:

```
run_greml.sh
```

Step 1: Slurm driver and fanout

The usual entry point is:

```
sbatch run_greml.sh
```

If no CLI arguments are supplied, 'run_greml.sh' defaults to fanout mode and submits 12 child jobs:

```
2 SNP sets x 2 expression sources x 3 normalization modes = 12 runs
```

Default fanout dimensions:

```
SNP sets: hm3_no_mhc, all_snps
Expression sources: tpm, tmm
Normalization types: irnt, inverse_normal, raw
```

Current child run labels look like:

```
all_snps_tmm_raw_peerauto_pmg0_npc5
all_snps_tmm_irnt_peerauto_pmg0_npc5
hm3_no_mhc_tpm_inverse_normal_peerauto_pmg0_npc5
```

The script prints a combo hash, but the current run directory name does not include that hash.

Step 2: FILTER_EUROPEANS

This process:

1. Reads GEUVADIS SDRF metadata.
2. Detects the population column by searching for 'ancestry category'.
3. Detects the RNA run column by searching for 'ENA_RUN'.
4. Keeps European groups:

```
British
Finnish
Tuscan
Utah
```

5. Maps GEUVADIS samples to PLINK '.fam' IIDs.
6. Optionally intersects with 'unrelated_keep_file'.
7. Emits:

```
eur_keep.txt
eur_map.tsv
```

'eur_keep.txt' is the FID/IID keep list used by GCTA. 'eur_map.tsv' maps RNA run IDs to genotype FID/IID and is used by phenotype preparation.

Step 3: GENERATE_PCA

This process builds the GRM:

```
gcta --bfile <plink prefix> \
  --keep eur_keep.txt \
  <optional --extract hm3_no_mhc_snps.txt> \
  --make-grm \
  --out genotype_grm \
  --thread-num <cpus>
```

Then it runs PCA from the GRM:

```
gcta --grm genotype_grm \
  --pca 5 \
  --out geno_pca \
  --thread-num <cpus>
```

Outputs are published to:

```
results/pca/<runLabel>/
```

Key outputs:

```
geno_pca.eigenvec  
genotype_grm.grm.bin  
genotype_grm.grm.N.bin  
genotype_grm.grm.id  
genotype_grm.log
```

Scientific rationale:

- * The GRM defines expected genetic covariance among individuals.
- * PCA captures broad ancestry structure and is included as a fixed covariate.
- * HM3/no-MHC reduces complexity from high-LD and difficult regions and improves comparability with common SNP reference analyses.

Step 4: PREPARE_PHENOTYPES

This process runs:

```
nf/bin/prepare_phenotypes.R
```

It receives:

```
TPM matrix  
count matrix  
eur_map.tsv  
geno_pca.eigenvec  
PLINK .fam  
phenotype prefix  
PEER settings  
GTEx-style filter thresholds  
expression source  
normalization type
```

The script performs:

1. Sample mapping and ordering.
2. Gene ID cleanup.
3. Duplicate gene handling.
4. Mandatory expression filter.
5. TMM or TPM phenotype-source selection.
6. RAW or inverse-normal phenotype normalization.
7. PEER factor estimation.
8. PCA and PEER covariate merge.
9. GCTA-compatible phenotype and qcovar file writing.

Outputs are published to:

```
results/data/
```

Important files:

```
pheno_<runLabel>.phenotypes.tsv  
pheno_<runLabel>.qcovar  
pheno_<runLabel>.gene_index_map.txt  
pheno_<runLabel>.filtered_gene_ids.txt  
pheno_<runLabel>.phenotypes.tsv.gz
```

The phenotype and qcovar files intentionally have no header because GCTA expects that format. The gene index map has a header.

Step 5: ESTIMATE_HERITABILITY

This process runs once per gene. The workflow reads the gene index map, then submits one 'ESTIMATE_HERITABILITY' task per gene.

GREML command:

```
gcta --grm genotype_grm \
--pheno pheno_<runLabel>.phenotypes.tsv \
--mphen <gene index> \
--qcovar pheno_<runLabel>.qcovar \
--reml \
--out <gene>_greml \
--thread-num 1
```

The output '.hsq' file contains:

```
V(G)
V(e)
Vp
V(G)/Vp
Pval
```

The pipeline extracts:

```
h2_GREML
SE_GREML
Pval_GREML
```

HE regression is implemented but disabled by default:

```
run_he = false
```

This was disabled because earlier HE runs failed due an Intel MKL shared-library issue in the GCTA AppRun environment. GREML is the primary output.

Step 6: SUMMARIZE_RESULTS

This process concatenates one '.stats' file per gene into:

```
results/summary/final_heritability_summary_<runLabel>.tsv
```

Columns:

```
Gene
Status
Index
h2_GREML
SE_GREML
Pval_GREML
h2_HE
SE_HE
```

Each gene gets 'PASS' or 'FAIL'. Per-gene failures do not necessarily fail the whole workflow because the workflow writes a '.stats' row and exits the gene job cleanly.

7. Parameter Table And Scientific Meaning

Sample and genotype parameters

```
eur_pops = ["British", "Finnish", "Tuscan", "Utah"]
```

Meaning:

- * Keeps the four European GEUVADIS populations.
- * Excludes YRI because the primary analysis is EUR-only.
- * Mimics GEUVADIS/Hernandez-style European analysis.

```
unrelated_keep_file = ""
```

Meaning:

- * Optional one-column IID or two-column FID/IID file.
- * When supplied, it restricts mapped GEUVADIS EUR samples to an external unrelated sample set.
- * This is the preferred way to mimic Hernandez-style cryptic-relatedness pruning.

```
use_hm3_no_hla = true
```

Meaning:

- * Uses HapMap3 SNPs excluding the MHC/HLA region when true.
- * Rationale: HM3 SNPs are a common well-imputed/common-variant reference set; excluding MHC reduces extreme LD and complex-region behavior.

```
n_pcs = 5
```

Meaning:

- * Uses the top five genotype PCs as covariates.
- * Rationale: genotype PCs capture broad ancestry structure even within EUR.
- * This is slightly different from GTEx V7, which listed top three genotype PCs, but five PCs is a conservative choice for a smaller population-genetic analysis.

Expression filtering parameters

```
gtex_tpm_threshold = 0.1
gtex_count_threshold = 6
gtex_sample_frac_threshold = 0.2
```

Meaning:

- * A gene must have TPM ≥ 0.1 in at least 20% of samples.
- * The same gene must also have count ≥ 6 in at least 20% of samples.
- * Both conditions must be true.

Scientific rationale:

- * TPM threshold removes near-background transcription.
- * Count threshold removes genes without enough read evidence.
- * The 20% rule prevents a gene expressed in only a few outlier individuals from entering the GREML analysis.
- * This follows GTEx-style eQTL expression filtering.

Expression-source parameters

```
expression_source = "tmm"
```

Allowed:

```
tmm
tpm
```

If 'tmm', the pipeline starts from counts, calculates TMM factors with edgeR, and uses TMM-normalized CPM.

If 'tpm', the pipeline uses Salmon-derived TPM directly.

Scientific difference:

- * TPM corrects within sample for gene/transcript length and sequencing depth.
- * TMM uses raw count composition across samples to estimate effective library size scaling factors.

* TPM is useful for abundance; TMM is often preferred for cross-sample count comparisons because it addresses RNA composition bias.

Normalization parameters

```
normalization_type = "irnt"
```

Allowed:

```
raw  
irnt  
inverse_normal
```

'raw':

- * Uses TMM CPM or TPM values directly.
- * Most biologically interpretable scale.
- * More vulnerable to zero inflation and skewness.

'irnt':

- * Uses ' $\log_2(x + 1)$ '.
- * Applies per-gene inverse-normal transform with Blom offset ' $3/8$ '.
- * Makes each gene approximately standard-normal across samples.
- * Less interpretable in expression units but better aligned with linear-model assumptions.

'inverse_normal':

- * Similar to 'irnt', but uses rank offset ' 0.5 '.
- * Kept mainly for comparison.

PEER parameters

```
peer_nk = "auto"  
peer_max_genes = 0
```

GTEX-style auto rule:

```
N < 150: 15 PEER factors  
150 <= N < 250: 30 PEER factors  
250 <= N < 350: 45 PEER factors  
N >= 350: 60 PEER factors
```

Scientific rationale:

- * PEER estimates hidden factors that capture technical and biological unwanted variation, such as batch effects or broad latent expression patterns.
- * GTEx chose the sample-size rule based on optimizing eGene discovery.
- * 'peer_max_genes = 0' means use all filtered genes for PEER.
- * Setting 'peer_max_genes > 0' uses top-variance genes only and is a speed option.

GCTA parameters

```
run_he = false
```

Meaning:

- * Only GREML is run by default.
- * Haseman-Elston regression is optional and currently skipped.

```
gcta_path = /gpfs/data/mostafavilab/programs/GCTA/...
```

Meaning:

* Exact GCTA executable used by the workflow.

8. How To Run The Workflow

Go to the project root on HPC:

```
cd /gpfs/data/mostafavilab/sool/analysis/GeneExpression/20260428_GE_GEUVDIS_v2/GeneExpression
```

Submit the default 12-run fanout:

```
sbatch run_greml.sh
```

Submit a single run:

```
GREML_FANOUT=0 sbatch run_greml.sh \  
--use_hm3_no_hla false \  
--expression_source tmm \  
--normalization_type raw
```

Run with Hernandez-style unrelated keep list:

```
GREML_FANOUT=0 sbatch run_greml.sh \  
--use_hm3_no_hla false \  
--expression_source tmm \  
--normalization_type raw \  
--unrelated_keep_file /path/to/geuvadis_1000g_unrelated_eur.keep
```

Submit 12 fanout runs with unrelated keep file:

```
GREML_FANOUT=1 GREML_FANOUT_INTERVAL_SEC=0 sbatch run_greml.sh \  
--unrelated_keep_file /path/to/geuvadis_1000g_unrelated_eur.keep
```

Disable resume:

```
GREML_RESUME=0 sbatch run_greml.sh ...
```

Use a different run root:

```
GREML_RUN_DIR=/path/to/output_root sbatch run_greml.sh ...
```

Use a different scratch root:

```
NXF_WORK_ROOT=/gpfs/scratch/<user>/nextflow_work/greml_production sbatch run_greml.sh ...
```

9. Expected Run Directory Layout

A run directory looks like:

```
runs/all_snps_tmm_raw_peerauto_pmg0_npc5/  
  nextflow_logs/  
  results/  
    data/  
    pca/  
    summary/
```

Data folder:

```
results/data/pheno_all_snps_tmm_raw.phenotypes.tsv
results/data/pheno_all_snps_tmm_raw.phenotypes.tsv.gz
results/data/pheno_all_snps_tmm_raw.qcovar
results/data/pheno_all_snps_tmm_raw.gene_index_map.txt
results/data/pheno_all_snps_tmm_raw.filtered_gene_ids.txt
```

PCA/GRM folder:

```
results/pca/all_snps_tmm_raw/genotype_grm.grm.bin
results/pca/all_snps_tmm_raw/genotype_grm.grm.N.bin
results/pca/all_snps_tmm_raw/genotype_grm.grm.id
results/pca/all_snps_tmm_raw/geno_pca.eigenvec
```

Summary folder:

```
results/summary/final_heritability_summary_all_snps_tmm_raw.tsv
```

10. Quality Checks After Running

Check final summary exists:

```
ls runs/*_peerauto_pmg0_npc5/results/summary/final_heritability_summary*.tsv
```

Check PASS/FAIL counts:

```
awk -F'\t' 'NR>1 {n[$2]++} END{for (k in n) print k,n[k]}' \
runs/all_snps_tmm_raw_peerauto_pmg0_npc5/results/summary/final_heritability_summary*.tsv
```

Check sample count in GRM:

```
wc -l runs/all_snps_tmm_raw_peerauto_pmg0_npc5/results/pca/all_snps_tmm_raw/genotype_grm.grm.id
```

Check samples against PLINK 'fam':

```
PLINK_PREFIX=gpfs/data/mostafavilab/shared_data/LDSC_data/1000G_EUR_Phase3_plink/1000G.EUR.QC.22
RUN_DIR=gpfs/data/mostafavilab/sool/analysis/GeneExpression/20260428_GE_GEUVDIS_v2/GeneExpression/runs
/all_snps_tmm_raw_peerauto_pmg0_npc5
comm -23 \
<(awk '{print $2}' "$RUN_DIR/results/pca/all_snps_tmm_raw/genotype_grm.grm.id" | sort -u) \
<(awk '{print $2}' "${PLINK_PREFIX}.fam" | sort -u)
```

No output means all used samples are in the genotype panel.

11. Downstream Analysis Scripts And Plots

All diagnostic analyses write under:

```
runs/_analysis/
```

Global h2 summary boxplot

Script:

```
scripts/plot_greml_h2_summary_boxplot.R
```

Outputs:

```
runs/_analysis/plot_greml_h2_summary_boxplot/tables/summary_of_summaries.tsv
runs/_analysis/plot_greml_h2_summary_boxplot/plots/greml_h2_boxplot_by_setting.png
```

Scientific rationale:

- * Shows distribution of PASS 'h2_GREML' estimates across all settings.
- * Helps identify which preprocessing choices inflate, deflate, or destabilize h2.
- * Mean labels are useful because h2 distributions can be skewed.

Pairwise h2 comparisons

Script:

```
scripts/pairwise_h2_comparisons.R
```

Main plot categories:

```
SNP set: ALL vs HM3/no-MHC
Expression source: TMM vs TPM
Normalization: RAW vs IRNT
```

Scientific rationale:

- * Each scatter plot compares gene-by-gene h2 estimates between two settings.
- * Identity line shows perfect agreement.
- * Color by expression level tests whether low expression drives disagreement.
- * Color by Wald Z tests whether discrepancies are concentrated among weak or strong h2 estimates.
- * Discrepancy counts quantify genes where one method gives high h2 and another gives near-zero h2.

Expression decile h2 analysis

Script:

```
scripts/tmm_expression_h2_deciles.R
```

Outputs:

```
runs/_analysis/tmm_expression_h2_deciles/plots/tmm_expression_h2_deciles_boxplot.png
runs/_analysis/tmm_expression_h2_deciles/plots/median_tpm_expression_h2_deciles_boxplot.png
runs/_analysis/tmm_expression_h2_deciles/plots/expression_skewness_by_decile_boxplot.png
```

Scientific rationale:

- * Tests whether gene-level h2 is higher for more highly expressed genes.
- * Uses TMM RAW mean expression and TPM RAW median expression as abundance bins.
- * Filters very low expression genes to avoid bins dominated by near-zero signals.
- * Adds skewness by expression bin to check whether distribution shape changes across abundance strata.

Default thresholds:

```
MIN_LOG2_MEAN_TMM_PLUS1 = 1.5
MIN_LOG2_MEDIAN_TPM_PLUS1 = 0
```

TMM raw skewness analysis

Script:

```
scripts/tmm_raw_skewness_analysis.R
```

Outputs include:

```
tmm_raw_skewness_hist_density_<METHOD>.png
tmm_raw_skewness_mean_vs_skew_<METHOD>.png
tmm_raw_skewness_selected_gene_distributions_<METHOD>.png
tmm_vs_tpm_h2_discrepancy_expr_distributions_<METHOD>.png
all_raw_h2_vs_hm3_raw_h2_tmm_colored_by_log2meanexpr_cap*.png
all_raw_h2_vs_hm3_raw_h2_tmm_colored_by_skewness_cap*.png
```

Scientific rationale:

- * Skewness detects whether raw expression phenotypes are dominated by long right tails or zero inflation.
- * Selected gene histograms give concrete examples for lab discussion.
- * H2 discrepancy examples test whether TPM/TMM disagreement is visible at the expression distribution level.
- * ALL-vs-HM3 colored scatter tests whether SNP-set sensitivity is linked to expression abundance or skewness.

TPM vs TMM phenotype correlations

Script:

```
scripts/tpm_tmm_peer_correlations.R
```

Outputs:

```
tpm_tmm_peer_gene_correlations.tsv
tpm_tmm_peer_correlation_summary.tsv
tpm_tmm_peer_correlation_boxplot.png
tpm_tmm_peer_mean_scatter.png
```

Scientific rationale:

- * Compares TPM-derived and TMM-derived phenotype values directly.
- * High per-gene correlation means expression source choice is unlikely to change that gene's phenotype ranking across individuals.
- * Low correlation identifies genes where TPM/TMM choice may alter h2.

Regulatory/repeat downstream analysis

Script:

```
scripts/downstream_h2_regulatory_repeat_analysis.R
```

This is a separate biological follow-up pipeline. It links h2 to:

- * 's_het post_mean'
- * gene regulatory burden
- * enhancer/open chromatin overlap
- * repeat classes/families/subtypes
- * randomized repeat background

Scientific rationale:

- * Tests whether gene expression heritability varies with evolutionary constraint.
- * Tests whether regulatory architecture or repeat burden differs across gene bins.
- * Observed-vs-background repeat analysis asks whether repeat patterns exceed random genomic expectation.

12. Interpretation Guide For Main Plots

h2 boxplot by setting

Use this first. It answers:

- * Which pipeline choices produce higher h2?
- * Are distributions broad, compressed, or outlier-heavy?
- * Are PASS counts similar across methods?

Do not interpret differences without checking PASS/FAIL counts.

Pairwise h2 scatter plots

Use these to diagnose gene-level stability.

Interpretation:

- * Points on diagonal: stable gene h2 across methods.
- * Points above diagonal: y-axis method estimates higher h2.
- * Points below diagonal: x-axis method estimates higher h2.
- * Low-expression colored outliers: likely expression-scale or low-information issue.
- * High-Z outliers: more likely biologically meaningful or at least statistically stable.

Expression decile h2 plots

Use these to ask whether expression abundance drives h2.

Interpretation:

- * Increasing h2 across deciles: low-expression genes may suppress h2 estimates.
- * Flat h2 across deciles: h2 estimates are robust to abundance after filtering.
- * Mean greater than median: h2 distribution is right-skewed in that bin.

Skewness plots

Use these to ask whether raw expression distributions are suitable for GREML.

Interpretation:

- * High positive skew: a few individuals have much higher expression.
- * High zero fraction: raw-scale h2 may be unstable.
- * IRNT should reduce sensitivity to skew because it rank-normalizes per gene.

TPM/TMM correlation plots

Use these to ask whether expression source changes phenotype rankings.

Interpretation:

- * High correlation: TPM and TMM mostly preserve individual-level ordering.
- * Low correlation: TPM/TMM choice can change the phenotype and therefore h2.

13. Known Caveats

1. The default genotype prefix ends with '.22'. Confirm whether the analysis is intentionally chromosome 22 only or whether genome-wide genotype files should be merged or supplied.
2. 'params.grm_prefix' exists but the code currently uses hard-coded 'genotype_grm'.
3. 'run_he=true' may fail unless the GCTA AppRun MKL runtime issue is fixed.

The current workflow intentionally sets 'run_he=false'.

4. Full reuse mode skips sample filtering and phenotype preparation. If 'unrelated_keep_file' is set while reusing GRM and phenotype files, the keep file does not change those reused artifacts.

5. Phenotype-only reuse with a newly generated GRM can mismatch normalization and sample set. The workflow warns but does not stop.

6. Per-gene GREML failures do not necessarily fail the whole workflow. Always inspect 'Status' in the summary table.

7. Several downstream scripts have hard-coded HPC paths. Before rerunning on a new project copy, inspect the first 20–40 lines of each R script.

8. Plot files are generated on the HPC run directory under 'runs/_analysis/'; they are not stored in this local repo by default.

14. Minimal Handoff Checklist

Before a new analyst reruns the workflow:

1. Confirm the genotype prefix is correct and genome-wide if needed.
2. Confirm the GEUVADIS TPM/count matrices exist.
3. Confirm 'gcta_path' works on compute nodes.
4. Confirm the 'r_peer' environment has 'data.table', 'edgeR', 'peer', and ideally 'matrixStats'.
5. Decide whether to provide 'unrelated_keep_file'.
6. Decide whether to run all 12 settings or a single setting.
7. Keep 'run_he=false' unless the MKL issue is solved.
8. Run a single small/default job first if changing input paths.
9. Check sample count in 'genotype_grm.grm.id'.
10. Check PASS/FAIL counts in the final summary.
11. Run diagnostic plots under 'scripts/'.
12. Archive both 'nextflow_logs/' and 'results/'.

15. Source List

GEUVADIS/1000G:

* International Genome Sample Resource GEUVADIS data collection:

<https://www.internationalgenome.org/data-portal/data-collections/geuvadis.html>

* GEUVADIS project summary: RNA-seq from 1000 Genomes samples across CEU, FIN, GBR, TSI, and YRI.

Salmon:

* Patro et al. Salmon provides fast and bias-aware quantification of transcript expression. Nature Methods 2017. <https://www.nature.com/articles/nmeth.4197>

TMM:

* Robinson and Oshlack. A scaling normalization method for differential expression analysis of RNA-seq data. Genome Biology 2010. <https://genomebiology.biomedcentral.com/articles/10.1186/gb-2010-11-3-r25>

GTEx methods:

* GTEx Analysis Methods V7 PDF.

https://storage.googleapis.com/gtex-public-data/Portal_Analysis_Methods_v7_09052017.pdf

GCTA/GREML:

* Yang et al. GCTA: A Tool for Genome-wide Complex Trait Analysis. American Journal of Human Genetics 2011. <https://www.sciencedirect.com/science/article/pii/S0002929710005987>

* GCTA documentation and notes on GRM and relatedness pruning:
<https://yanglab.westlake.edu.cn/software/gcta/>

GEUVADIS expression heritability:

* Hernandez et al. Ultrarare variants drive substantial cis heritability of human gene expression. Nature Genetics 2019. <https://www.nature.com/articles/s41588-019-0487-7>

* Dahl et al. On Negative Heritability and Negative Estimates of Heritability. Genetics 2020. <https://academic.oup.com/genetics/article/215/2/343/5930411>